

Yutao Jin

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Research Area: Computer Vision (Semantic Segmentation), Time Series (Prediction, Anomaly Detection)

Education

Double Bachelor's Degree Program

Sep. 2022 – Jun. 2026

- *Tianjin University of Technology, B.Eng. in Computer Science*

Tianjin, China

- *University of Quebec, B.Eng. in Business Intelligence*

Chicoutimi, Canada

- Weighted Average Mark: 91.91/100; rank: 3/174.
- Undergraduate thesis: *Multimodal Data-Driven Approaches for Surface Crack Detection*.
- English Proficiency: IELTS Academic Overall Band Score 6.5.

Research Experience

Key Laboratory of Computer Vision and System, MOE

Tianjin, China

Undergraduate Researcher, Efficient Representation Learning and Structure-Aware Modeling for Crack Segmentation

Apr. 2025 – Present

- Developed lightweight and structure-aware crack segmentation models for complex civil infrastructure images, targeting thin, low-contrast, discontinuous, and texture-confused crack patterns in real-world inspection scenarios.
- Explored edge-guided, frequency-aware, and multi-directional representation learning to improve crack topology preservation, boundary continuity, and long-range structural modeling under computational constraints.
- Designed topology-preserving decoding and multi-scale feature interaction modules to strengthen fine-grained structural consistency and improve robustness across heterogeneous crack patterns.
- Extended the work to quality-aware RGB-X crack segmentation, studying reliable fusion of RGB, depth, thermal, and polarization modalities under incomplete or corrupted sensing conditions.
- Contributed to multiple crack segmentation manuscripts, including accepted work in *Engineering Structures* and ICIC 2025, and ongoing work on RGB-X fusion and Mamba-based lightweight segmentation.

Technical University of Denmark

Kongens Lyngby, Denmark

Undergraduate Researcher, Efficient Spatio-Temporal Wind Forecasting

Jan. 2025 – Apr. 2025

- Studied efficient spatio-temporal learning for multi-horizon wind power forecasting, focusing on resource-constrained prediction in real-world renewable energy systems.
- Modeled temporal dynamics and inter-turbine relationships through similarity-aware interaction learning and attention-based feature fusion, reducing redundant representations while preserving forecasting capacity.
- Contributed to *LightWind*, a lightweight wind power forecasting framework accepted by *IEEE Sensors Journal*.

Engineering Research Center of Learning-Based Intelligent Systems, MOE

Tianjin, China

Undergraduate Researcher, Time Series Modeling and Anomaly Detection

Jun. 2024 – Jan. 2025

- Explored frequency-aware representation learning for log anomaly detection, aiming to separate low-frequency behavioral shifts from high-frequency transient deviations in noisy and data-scarce event streams.
- Developed lightweight and parameter-efficient architectures for large-scale monitoring systems, with emphasis on practical deployment under high-throughput and limited-resource conditions.
- Investigated factorized temporal and channel mixing for multivariate industrial time-series data, especially SCADA-based wind turbine fault diagnosis with imbalanced fault distributions.
- Studied multi-scale temporal dynamics and class imbalance to improve generalization across heterogeneous fault modes and operational environments.

Publications and Manuscripts

1. **Y. Jin**, H. Liu, et al.. “Crack segmentation in civil infrastructure using conditional modulation and hierarchical dual path networks.” *Engineering Structures*. **Accepted**. JCR Q1, IF: 7.62. DOI: 10.1016/j.engstruct.2025.121162.
2. **Y. Jin**, M. Liu, et al.. “LightWind: A Lightweight Framework for Efficient Wind Power Forecasting.” *IEEE Sensors Journal*. **Accepted**. JCR Q1, IF: 4.65. DOI: 10.1109/JSEN.2025.3600483.
3. **Y. Jin**, H. Liu, et al.. “EMCrossNet: Edge-Guided Bidirectional Mamba Fusion Network for Crack Segmentation.” *International Conference on Intelligent Computing*. ICIC 2025, **Accepted**. DOI: 10.1007/978-981-96-9869-1_12.
4. **Y. Jin**, X. Cheng.. “TRACER-X: Topology-Preserving Hierarchical Fusion for Quality-Aware RGB-X Crack Segmentation.” *The Forty-First AAAI Conference on Artificial Intelligence*. AAAI 2027, **Under Review**.
5. **Y. Jin**, H. Liu, et al.. “FracturaMamba: Lightweight and Efficient Crack Segmentation for Autonomous Robotic Inspection in Resource-Constrained Environments.” *Engineering Structures*. **Under Review**. JCR Q1, IF: 7.62.
6. M. Liu, **Y. Jin**, et al.. “A Factorized Temporal and Channel Mixing Model for Wind Turbine Fault Diagnosis with Imbalanced SCADA Data.” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*. **Under Review**. JCR Q1, IF: 8.1.
7. **Y. Jin**, C. Jia, et al.. “LogSPECTRA: Learning Spectral Baseline–Perturbation Dynamics in Data-Scarce Semi-Structured Event Streams.” *Manuscript*. In preparation.

Honors and Awards

- National Second Prize, Main Track, 18th Challenge Cup National Competition of Extracurricular Academic and Technological Works, 2025.
- National Second Prize, Special Application Track, 18th Challenge Cup National Competition of Extracurricular Academic and Technological Works, 2025.
- National Third Prize, China International College Students’ Innovation Competition, 2025.
- Second Prize, North China Regional Competition, Network Technology Challenge, 2025.
- Third Prize, National Finals of the 6th Global Artificial Intelligence Algorithm Competition, 2024.
- People’s Scholarship, University-Level, Tianjin University of Technology, 2024.
- People’s Scholarship, University-Level, Tianjin University of Technology, 2023.

Research Profile

- Primary research direction: efficient and robust machine learning for safety-critical perception and industrial monitoring.
- Domain experience: civil infrastructure crack segmentation, RGB-X multimodal fusion, wind power forecasting, wind turbine fault diagnosis, log anomaly detection, and SCADA-based time-series modeling.
- Methodological focus: lightweight architectures, state space models, frequency-aware representation learning, spatio-temporal modeling, multi-scale feature interaction, topology preservation, and imbalanced data learning.